

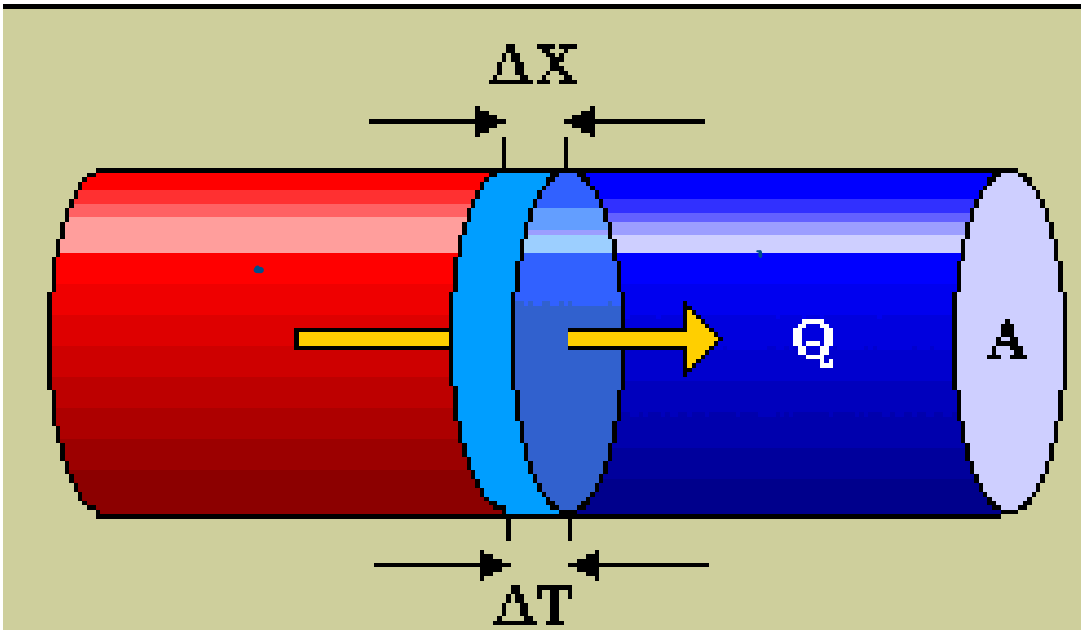
STM Exercise 4

➤ Thermal Properties

Thermal conductivity

Materials attitude to transfer thermal energy from one system at higher Temperature to another one at lower Temperature. (phonons)

Coefficient of thermal conductivity (k , W/mK) : energy (Q , J) flowing in one second (t , 1 second) through one surface (A , 1 m^2), one meter thick (Δx , 1 m), when the ΔT difference between the two surfaces is 1 K.



$$\frac{Q}{A.t} = k \frac{\Delta T}{\Delta x}$$

Define the coefficient of thermal conductivity (k), its units and list k of polymers, metal, and ceramics. Explain the list order.

Calculate the amount of energy (in Joule) transmitted, in a first approximation, in 1 hour, through a 1 mm² polymer component, 2 mm thick, when in contact with water at 50 °C on one side and air at 20 °C on the other side. ($k=0,58$ W/mK) (write the number, do not write Joule)

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Coefficient of thermal conductivity

k (W/m K)

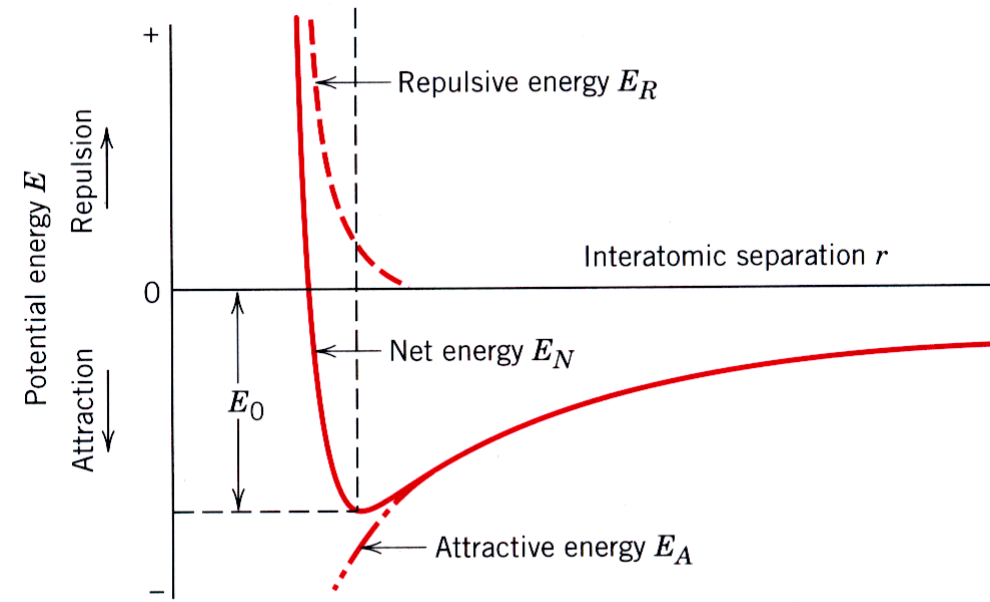
Al	247	Al ₂ O ₃	30	polyethylene	0.4
Cu	398	MgO	38	nylon	0.24
Fe	81	SiO ₂ (glass)	2	PTFE	0.25
Steel	52	soda-lime glass	1.7		
Stainless steel	16	MgAl ₂ O ₄	15		

$$k_{\text{metals}} > k_{\text{ceramics}} \gg k_{\text{polymers}}$$

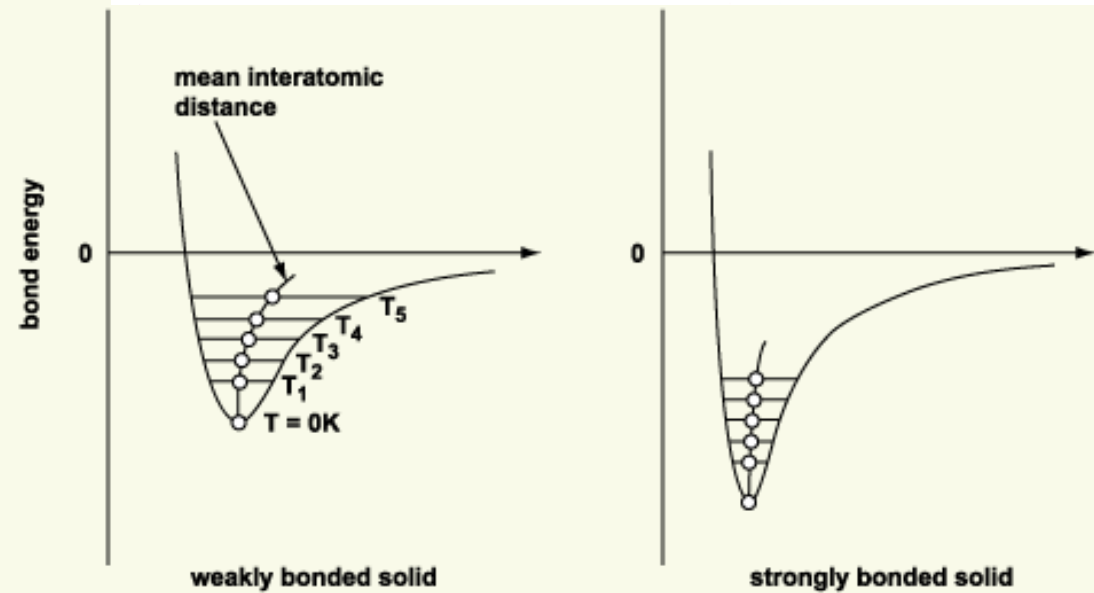
CTE and bond strength

Strongly bonded solids,
high E ,
high melt T

Lower α



(b)



Define the coefficient of thermal expansion (CTE), its units and list the CTE of polymers, metal, glasses and ceramics. Explain the list order.

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Linear expansion

$$\frac{\Delta L}{L} = \alpha_L \Delta T$$

- Polymers: $100 \times 10^{-6} \text{ K}^{-1}$
- Metal: $15\text{-}30 \times 10^{-6} \text{ K}^{-1}$
- Glasses: $0,5\text{-}20 \times 10^{-6} \text{ K}^{-1}$
- Ceramics: $3\text{-}7 \times 10^{-6} \text{ K}^{-1}$

The relative expansion (also called strain) divided by the change in temperature is called the material's **coefficient of linear thermal expansion (CTE)**

CTE of different materials

α Ceramics < α metals << α polymers

α silica glass = $0.5 \times 10^{-6} \text{ K}^{-1}$

Materiale	X ($10^{-6} \text{ }^{\circ}\text{K}^{-1}$)
Al	25
Cu	16.6
Fe	12
Pb	29
Mg	25
Ni	13
Si	3
Ti	8.5
W	4.5
Steel	12
Stainless steel	17.3
Brass (Cu-Zn)	18.9
Invar (Fe-36% Ni)	1.54
Polietilene	100
Polistirene	70
Resina epossidica	55
6,6-Nylon	80
Amorphous silica	0.55
Al_2O_3	6.7
Si_3N_4	3.3
SiC	4.3

α glasses = $0.5 - 20 \times 10^{-6} \text{ K}^{-1}$

Thermal expansion coefficient

- SiC = $4.3 \times 10^{-6} / \text{K}$ (20 - 1000°C)
- Al_2O_3 = $8.6 \times 10^{-6} / \text{K}$ (0 - 1000°C)
- Si_3N_4 = $3.2 \times 10^{-6} / \text{K}$ (25 - 1225°C)
- Comparison with metals
 - Cu = $\sim 18 \times 10^{-6} / \text{K}$ (20 - 300°C)
 - Al = $\sim 26 \times 10^{-6} / \text{K}$ (20 - 300°C)
 - carbon steel = $\sim 13 \times 10^{-6} / \text{K}$ (20 - 400°C)

The CTE of a steel is $(14 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1})$: calculate the elongation of a steel rod of 50 m, when temperature increases from 20 to 50 $^{\circ}\text{C}$.

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$$\frac{\Delta L}{L} = \alpha_L \Delta T$$

Es 1.

An Al bar, 10 m long, is cooled from 38 to -10°C . Knowing that $\alpha=23.6 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$,
how does its length change?

Es 2. A steel bar is heated from 20 to 80°C with the ends constrained. Knowing that:

$$E = 207 \text{ GPa}$$

$$\alpha = 12 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

Determine the developed tension

Es 3. A brass bar constrained at the ends, and stress-free at 20°C, must be heated without exceeding a compressive stress of 172 MPa.

Up to what temperature can it be heated?

$E = 100 \text{ GPa}$

$\alpha = 20 \cdot 10^{-6} \text{ }^{\circ}\text{C}^{-1}$

Es 4. At what T must a cylindrical bar of W having $d = 10$ mm, and a stainless steel plate with a hole of $d=9.988$ mm, be heated to make the bar assume the same diameter as the hole?

$$T_1 = 25^\circ\text{C}$$

$$\alpha(\text{W}) = 4.5 \cdot 10^{-6} \text{ }^\circ\text{C}^{-1}$$

$$\alpha(\text{inox}) = 16 \cdot 10^{-6} \text{ }^\circ\text{C}^{-1}$$

Es 6.

Calculate the coefficient of thermal conductivity k of copper knowing that from the two sides of a sheet 5 mm thick and with a square surface equal to $10 \times 10 \text{ cm}^2$ the temperatures are 20°C and 90°C and that in 5 minutes 16716 kJ are transmitted.